Model Program Book

LONG-TERM INTERNSHIP (On-Site/Virtual)

Designed & Developed by

ANDHRA PRADESH
STATE COUNCIL OF HIGHER EDUCATION

(STATUTORY BODY OF GOVERNMENT OF ANDHRAPRADESH)

PROGRAM BOOK FOR

LONG-TERM INTERNSHIP

(Onsite/Virtual)

Name of the Student: **Medidhi Naga Madhuri**

Name of the College: **Vignan Degree and Pg College**

Registration Number: **Y233082092**

Period of Internship: From: **16-12-2025** To: **14-03-2026**

Name & Address of the Intern Organization : **Smartinternz, SmartBridge Educational Services Pvt. Ltd.**

YEAR

2025-2026

An Internship Report
on

Artificial Intelligence and Machine Learning (AIML)

Submitted in accordance with the requirement for the degree of

Bachelor

In

Computer Science

Submitted by

Medidhi Naga Madhuri
(Y233082092)

Under the Faculty Guideship of

K. Pradeep

Assistant Professor

Department of Computer Science and Engineering

VIGNAN'S

Foundation for Science, Technology & Research

(Deemed to be UNIVERSITY)

-Estd. u/s 3 of UGC Act 1956

Student's Declaration

I, **Medidhi Naga Madhuri,** a student **Bachelor of Computer Science**, Regd.No. **Y233082092** of the Department **Computer Science , Vignan Degree and Pg College**, do hereby declare that I have completed the mandatory internship in **Artificial Intelligence and Machine Learning** from 16th December, 2025 to 20th February, 2026 in **SmartInternz** under the Faculty Guideship of **K. Pradeep**, Department of **Computer Science , Vignan Degree and Pg college.**

(Signature and Date)

Faculty Guide

K. Pradeep, M. Tech.

Head of the Department

Dr. K. Venkateswara Rao, Ph.D.

External Examiner

Acknowledgements

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of people whose ceaseless cooperation made it possible, whose constant guidance and encouragement crown all efforts with success.

I am grateful to **K. Pradeep, M. Tech.,** Assistant Professor, Department of Computer Science and Engineering for guiding through this project and for encouraging right from the beginning of the project till successful completion of the project. Every interaction with her was an inspiration.

I thank **Dr. K. Venkateswara Rao, Ph.D.**, Professor & HoD, Department of Computer Science and Engineering for support and Valuable suggestions

I also express our thanks to **Dr. K. Phaneendra Kumar**, Principal, Vignan's Lara Institute of Technology & Science for providing the resources to carry out the project.

I also express my sincere thanks to our Chairman **Dr. Lavu Rathaiah** for providing support and stimulating the environment for developing the project.

I also place my floral gratitude to all other teaching and lab technicians for their constant support and advice throughout this Virtual Internship.

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SMARTBRIDGE

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A Statutory Body of the Government of A.P)

This is to certify that Ms./Mr. **Medidhi Naga Madhuri** of **Bachelor of Computer Science** with Registered Hall ticket no. **Y233082092** under **Vignan Degree and Pg College, Pedapalakaluru, Guntur PIN-522009** of **ANU** has successfully completed Long-Term Internship of 240 hours (6 months) on **Artificial Intelligence & Machine Learning** Organized by **SmartBridge Educational Services Pvt. Ltd.** in collaboration with **Andhra Pradesh State Council of Higher Education.**

Certi&cate ID: **EXT-APSCHE_AIML-89571**

Date: 20-Mar-2026

Place: Virtual

Amarendar Katkam

Founder & CEO

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Special thanks to SmartBridge for mentoring me in Artificial Intelligence and Machine Learning and helping me build my skills in technologies such as Python, Machine Learning algorithms, Deep Learning, and data analysis. The hands-on learning opportunities and industry-oriented training I received were instrumental in enhancing my technical competence. Their expert insights into the latest AI/ML trends and best practices played a crucial role in shaping my project work and overall internship experience. I am truly grateful for their commitment and for providing a strong foundation that will benefit my future career.

Introduction

This report provides an extensive account of my two-month Artificial Intelligence and Machine Learning internship at SmartBridge Educational Services Pvt. Ltd., conducted in collaboration with the Andhra Pradesh State Council of Higher Education. The internship was designed to immerse students in cutting-edge industry practices while bridging the gap between academic learning and real-world AI/ML applications. My engagement with SmartBridge not only exposed me to advanced algorithms, data-driven methodologies, and analytical tools but also instilled in me an appreciation for problem-solving, model development, and iterative learning approaches.

Throughout the program, I gained in-depth experience in building and deploying machine learning models using Python, along with libraries and frameworks such as NumPy, Pandas, Scikit-learn, and TensorFlow/PyTorch. Each phase of the internship fostered holistic learning, starting from foundational concepts of data preprocessing and exploratory data analysis to hands-on assignments, model training, evaluation, and optimization. I was actively involved in working with datasets, implementing supervised and unsupervised learning algorithms, and developing predictive models. Regular mentorship from industry experts and interactive sessions enhanced my understanding of model selection, feature engineering, hyperparameter tuning, and performance evaluation.

A significant highlight of this internship was the opportunity to work on real- world and team-based projects that simulated industry scenarios. Collaborating with peers, I participated in data collection, preprocessing, model building, and validation processes. These experiences helped me understand the importance of clean data, proper evaluation metrics, and continuous model improvement. Projects such as sentiment analysis, classification models, and basic deep learning applications challenged me to apply theoretical knowledge to practical use cases. I also gained exposure to tools for version control, experiment tracking, and deployment concepts for machine learning models.

The holistic approach adopted by SmartBridge, combining technical learning with analytical thinking and collaboration, prepared me for the evolving field of Artificial Intelligence and Machine Learning. My experience was further enriched by continuous feedback, project reviews, and knowledge-sharing sessions. This internship has been a transformative journey, significantly enhancing my technical skills, critical thinking, and confidence to pursue a career in AI/ML and data-driven technologies.

Company Overview

SmartBridge Educational Services Pvt. Ltd. is a pioneering educational technology company dedicated to transforming how students and professionals acquire industry-relevant skills in today's fast-paced technological landscape. Established with the vision to "Bridge the Gap" between traditional academia and the requirements of modern industries, SmartBridge focuses on experiential learning through internships, live projects, workshops, and certification training across various domains such as Full Stack Development, Artificial Intelligence, Data Science, and Cloud Computing.

By partnering with leading educational institutions, government bodies like the Andhra Pradesh State Council of Higher Education, and industry experts, SmartBridge ensures that learners receive mentoring from experienced professionals and exposure to current best practices. The company is committed to fostering a culture of innovation, encouraging problem-solving and collaborative learning environments where students can apply theoretical knowledge in real-world scenarios.

Andhra Pradesh State Council of Higher Education

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SmartBridge's programs are uniquely structured to include hands-on projects, practical assignments, and regular feedback, enabling participants not only to build strong technical skills but also to develop project management, teamwork, and communication abilities essential for workplace success. Through their various initiatives, SmartBridge has positively impacted thousands of learners, empowering them to become job-ready and to contribute effectively to the competitive technology sector.

The commitment of SmartBridge to continuous improvement and industry alignment makes it a trusted partner for universities and students aiming to improve employability and drive innovation. Ultimately, SmartBridge aspires to be a catalyst in shaping the next generation of technology leaders in India and beyond.

Internship Description

My two-month internship with SmartBridge Educational Services Pvt. Ltd. was structured to provide comprehensive exposure to the end-to-end lifecycle of Artificial Intelligence and Machine Learning projects, from problem identification to model deployment. The journey began with the Ideathon Phase, where we explored real-world challenges and proposed data-driven solutions. Through brainstorming sessions and dataset understanding, our team identified the problem of detecting fraudulent online payment transactions and defined clear objectives focused on accuracy, reliability, and real-time prediction.

During the ideation stage, emphasis was placed on analytical thinking and problem definition. We studied transaction data, identified patterns, and refined our approach to ensure feasibility and impact. This phase helped us establish key goals such as minimizing false positives and improving fraud detection performance. It also strengthened our ability to translate real-world problems into structured machine learning tasks.

The internship then progressed into the Data Analysis and Project Planning phase, where we conducted exploratory data analysis, handled missing values, and performed feature engineering. We designed a machine learning pipeline and selected an appropriate technology stack including Python, Pandas, NumPy, Scikit- learn, and XGBoost. Tools like Jupyter Notebook and Git supported experimentation, visualization, and version control, ensuring an organized and efficient workflow.

In the Development Phase, we focused on building and training the machine learning model. I contributed to preprocessing the data, splitting it into training and testing sets, and training the XGBoost classifier. The model was evaluated using metrics such as accuracy, precision, recall, and F1-score. Continuous tuning and iteration helped improve the model's performance, making it suitable for detecting fraudulent transactions effectively.

Finally, the project moved into the Deployment and Testing Phase, where the trained model was integrated into a Flask-based web application for real-time predictions. We conducted functional and performance testing to ensure reliability and speed. The internship concluded with a final demonstration, showcasing a fully functional fraud detection system. This experience significantly enhanced my understanding of AI/ML workflows and strengthened my skills in data analysis, model development, and practical problem-solving.

Project Highlights

1. Ideation Phase

The internship began with a comprehensive project orientation at SmartBridge, aimed at bridging theoretical knowledge with real-world Artificial Intelligence and Machine Learning practices. During the Ideation Phase, our team focused on understanding the problem domain of online payment fraud detection. We explored transaction data scenarios and discussed the growing challenges in identifying fraudulent activities in digital payment systems.

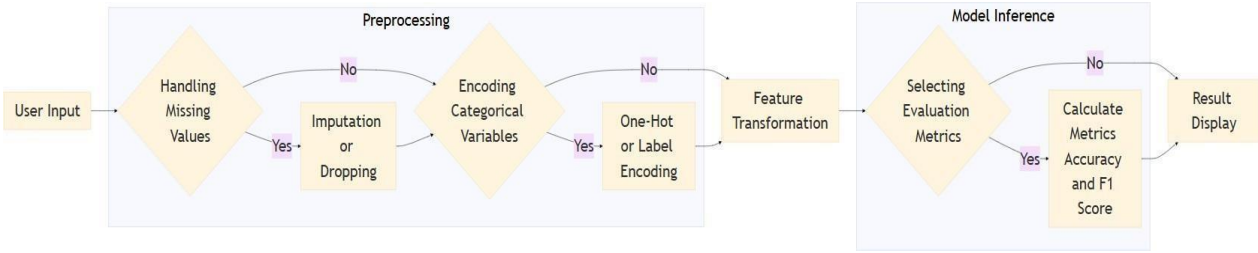
To gain deeper insights, we analyzed patterns in transaction behavior and identified key issues such as high false positives, evolving fraud techniques, and the need for real-time detection. Brainstorming sessions led to multiple ideas, including anomaly detection systems, rule-based filtering, and machine learning-based classification models. These discussions helped us evaluate different approaches to effectively tackle fraud detection.

After assessing feasibility, scalability, and impact, we finalized the approach of developing a machine learning-based fraud detection system using the XGBoost algorithm. The focus was on creating a solution that could accurately classify transactions as genuine or fraudulent while maintaining efficiency and reliability.

The ideation documentation included a well-defined problem statement, identified challenges, possible solution approaches, and justification for selecting the final model. This structured ideation process ensured that the project was aligned with real-world financial security needs and laid a strong foundation for subsequent phases of development.

2. Requirement Analysis

The requirement analysis phase translated our project idea into clearly defined objectives from both technical and system perspectives. We outlined functional requirements such as accepting transaction inputs, preprocessing data, generating fraud predictions, and displaying results to the user. Non-functional requirements included high model accuracy, low latency for real-time prediction, system reliability, data security, and scalability to handle large volumes of transactions.



A workflow diagram was created to represent the end-to-end pipeline, starting from user input to final prediction output. This included stages like data input, preprocessing, feature transformation, model inference, and result display. We also identified key decision points such as handling missing values, encoding categorical variables, and selecting appropriate evaluation metrics. This structured flow ensured smooth interaction between different components of the system.

To better understand data movement, we designed data flow diagrams representing the overall process of the system. These diagrams illustrated how transaction data flows from user input through various stages such as preprocessing, feature transformation, model inference, and final prediction output. Key steps like handling missing values, encoding categorical variables, loading the trained model, and generating predictions were clearly mapped. This helped in visualizing how data is processed and transformed at each stage of the machine learning pipeline, ensuring clarity in system design and implementation.

The system components and technology stack were defined as follows:

S.No	Component	Description	Technology
1	User Interface	Accepts transaction details and displays prediction results	HTML, CSS, Flask
2	Application Logic	Handles preprocessing, model loading, and prediction logic	Python
3	Machine Learning Model	Classifies transactions as fraud or genuine	XGBoost
4	Data Processing	Data cleaning, feature engineering, and transformation	Pandas, NumPy
5	Model Evaluation	Performance measurement and validation	Scikit-learn, Matplotlib
6	Deployment	Hosts the application for real-time usage	Flask (Local/Cloud Deployment)

The technology stack was selected based on efficiency and suitability for machine learning workflows. Python served as the core language, with libraries like Pandas and NumPy for data handling, Scikit-learn for preprocessing and evaluation, and XGBoost for model training. Flask was used to integrate the model into a web application, while tools like Jupyter Notebook and Git supported experimentation and version control.

The system characteristics were defined as follows:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Utilizes Python-based ML ecosystem for model building and deployment	Python, Pandas, NumPy, Scikit-learn, XGBoost, Flask
2	Security Implementations	Input validation, restricted model access, and basic authentication mechanisms	Flask Sessions, Input Validation
3	Scalable Architecture	Modular pipeline separating preprocessing, model, and application layers	ML Pipeline Architecture
4	Availability	Can be deployed on local servers or cloud platforms for accessibility	Flask, Cloud Platforms (optional)
5	Performance	Optimized model ensures fast prediction and efficient handling of structured data	XGBoost, NumPy

3. Project Design

Project design efforts focused on developing an effective and reliable solution for detecting fraudulent online transactions using machine learning. The system was conceptualized to address challenges such as inaccurate fraud detection, delayed responses, and the need for scalable, real-time analysis. Emphasis was placed on building a model-driven solution that ensures accuracy, speed, and ease of use for end users.

The proposed solution was structured around three key aspects: efficient data preprocessing to handle raw transaction data, a robust machine learning model (XGBoost) for accurate classification, and a real-time prediction interface for instant results. These components worked together to ensure that the system could identify fraudulent patterns while minimizing false positives and negatives.

The solution architecture was designed to enable smooth interaction between system components. The frontend interface collects transaction details from the user, which are then processed by the backend Flask application. The backend handles preprocessing, loads the trained model, and performs prediction, while the results are displayed back to the user. This structured flow ensured seamless integration between data handling, model inference, and user interaction.

Additionally, the system design included a modular structure separating data preprocessing, model training, and prediction functionalities. This modularity allows easy updates, such as retraining the model with new data or improving preprocessing

techniques without affecting the entire system. The design approach ensured flexibility, scalability, and maintainability of the application.

Basic UI design focused on simplicity and usability, providing clear input fields for transaction details and easy-to-understand prediction outputs such as “Fraud Detected” or “Transaction is Safe.” This ensured that even non-technical users could interact with the system effectively, making the machine learning solution practical for real-world usage.

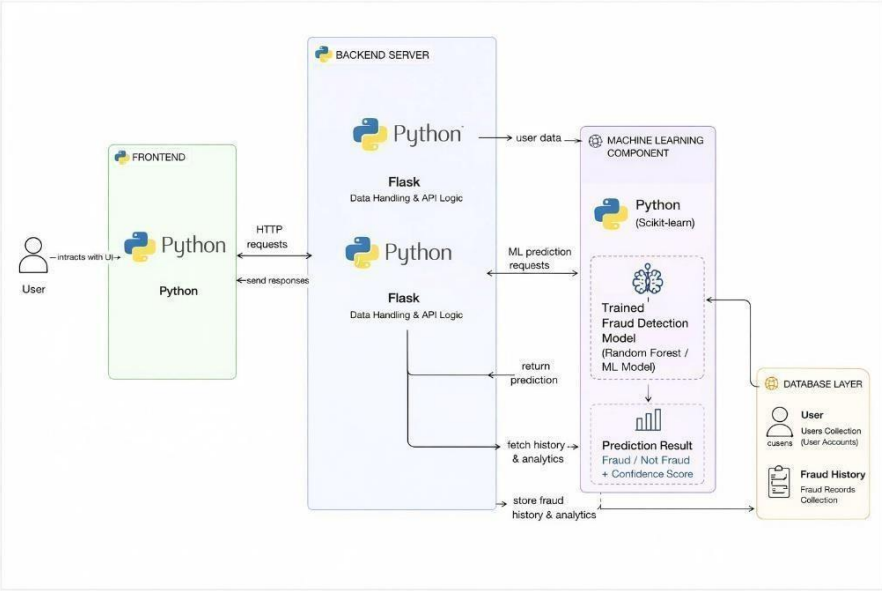


Fig: Solution Architecture

4. Performance Testing

Performance testing was a crucial phase to ensure that the Online Payments Fraud Detection System was not only accurate but also efficient, reliable, and capable of handling real-time predictions. The testing strategy focused on validating the model’s performance under different scenarios, including varying input data and multiple prediction requests, to ensure consistent behavior in practical usage.

Model evaluation played a key role in performance testing, where metrics such as accuracy, precision, recall, and F1-score were used to assess the effectiveness of the XGBoost model. Special attention was given to handling imbalanced data, ensuring that fraudulent transactions were correctly identified while minimizing false positives and false negatives. Confusion matrix analysis further helped in understanding classification performance.

System performance was also evaluated in terms of response time and efficiency. The application was tested to ensure that predictions were generated within a few seconds, making it suitable for real-time fraud detection. Multiple test cases were

ACTIVITY LOG FOR THE FIRST WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 16-12-2025	Orientation and overview of Fraud Detection System project	Understood project scope, objectives, methodology	
Day-2 17-12-2025	Teamforming and role assignment	Learned about team roles and collaboration and responsibility distribution	
Day-3 18-12-2025	Requirements gathering and initial brainstorming	Gained insights into requirement analysis and system design	
Day-4 19-12-2025	Empathy map and customer journey mapping	Learned design thinking and empathy mapping	
Day-5 20-12-2025	Problemstatement definition and idea finalization	Understood problem-solving approaches	
Day-6 22-12-2025	Finalized feature list and initial project planning	Learned project planning, task allocation, and scheduling basics	

WEEKLY REPORT

WEEK–1 (From 16-12-2025 to 22-12-2025)

Objective of the Activity Done:

To understand the fundamentals of the Fraud Detection System, analyze requirements, define the problem statement, and create a structured plan for project development.

Detailed Report:

The first week of the internship focused on understanding the Fraud Detection System using Flask and Machine Learning. It began with an orientation session explaining the project objectives, scope, and real-world applications in banking and online transactions. The team was formed, and roles were assigned to ensure smooth collaboration. Responsibilities such as backend development, machine learning implementation, and frontend design were clearly defined.

The team then worked on requirement analysis by studying existing fraud detection systems and identifying key features like transaction analysis and prediction accuracy. Empathy mapping helped in understanding user needs such as reliability and quick response. Based on this, the problem statement was finalized, and a basic project plan with core features was prepared, providing a clear roadmap for further development. Additionally, discussions were held to prioritize features based on feasibility and impact. This ensured that the project development would proceed in a structured and efficient manner.

ACTIVITY LOG FOR THE SECOND WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 23-12-2025	Created detailed product backlog and defined tasks for Fraud Detection System	Learned backlog creation and task prioritization	
Day-2 24-12-2025	Started sprint-1: Dataset collection and understanding	Understood sprint planning and data handling basics	
Day-3 26-12-2025	Performed data preprocessing and cleaning	Gained hands-on experience in missing values and data formatting	
Day-4 27-12-2025	Analyzed dataset and performed feature selection	Learned importance of selecting relevant features for model accuracy	
Day-5 29-12-2025	Implemented data visualization for pattern analysis	Understood trends and insights using graphs	
Day-6 30-12-2025	Sprint review and dataset validation	Improved skills in reviewing work and ensuring data quality	

WEEKLY REPORT

WEEK–2 (From 23-12-2025 to 30-12-2025)

Objective of the Activity Done:

To collect, analyze, and preprocess data for the Fraud Detection System.

Detailed Report:

During this week, the team focused on organizing the workflow by creating a detailed project backlog and initiating Sprint-1. The primary goal was to collect and understand the dataset required for the Fraud Detection System. The dataset was carefully analyzed to identify important attributes related to transaction patterns, which helped in building a strong base for further development.

As part of data preparation, preprocessing techniques were applied to clean and structure the dataset. Missing values were handled, duplicates were removed, and relevant features were selected to improve model performance. In addition, data visualization methods were used to analyze trends and patterns, enabling better understanding of fraudulent and non-fraudulent transactions.

By the end of the sprint, the dataset was properly cleaned and validated, making it ready for machine learning model development. The sprint review ensured data quality and consistency, while collaborative efforts within the team resulted in a well-organized and structured dataset, setting a solid foundation for the next phase of the project.

ACTIVITY LOG FOR THE THIRD WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 31-12-2025	Initiated machine learning model development for fraud detection	Learned basics of ML model implementation	
Day-2 02-01-2026	Implemented XGBoost algorithm for classification	Understood classification techniques	
Day-3 03-01-2026	Trained XGBoost model using processed dataset	Gained hands-on experience in model training	
Day-4 05-01-2026	Tested model performance and predictions	Learned evaluation of model accuracy	
Day-5 06-01-2026	Tuned XGBoost model parameters for better performance	Understood model optimization techniques	
Day-6 07-01-2026	Reviewed model results and finalized XGBoost model	Learned model evaluation and selection	

WEEKLY REPORT

WEEK–3 (From 31-12-2025 to 07-01-2026)

Objective of the Activity Done:

To develop, train, and evaluate machine learning models for Fraud Detection System.

Detailed Report:

During this week, the team focused on implementing machine learning models for fraud detection using the preprocessed dataset. The development began with understanding classification techniques and selecting suitable algorithms. The XGBoost algorithm was implemented due to its high performance and efficiency in handling structured data and imbalanced datasets.

The dataset was used to train the XGBoost model, and testing was carried out to evaluate its performance. The team analyzed metrics such as accuracy and prediction results to understand how effectively the model detects fraudulent transactions. This phase provided hands-on experience in model training, tuning, and improving performance.

By the end of the week, the XGBoost model was finalized based on its superior performance. The sprint review ensured that the model was reliable and ready for integration into the Flask application. This week played a key role in building the core intelligence of the system for accurate fraud detection.

ACTIVITY LOG FOR THE FOURTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 08-01-2026	Initiated Flask backend setup for fraud detection system	Understood Flask framework and project structure	
Day-2 09-01-2026	Developed routes for handling user input and prediction requests	Learned routing and request handling in Flask	
Day-3 10-01-2026	Integrated machine learning model with Flask backend	Understood model integration into web application	
Day-4 12-01-2026	Created input form for transaction details	Learned form handling and data collection	
Day-5 13-01-2026	Implemented prediction logic and result display	Learned how to show model output to users	
Day-6 17-01-2026	Tested backend functionality and fixed bugs	Improved debugging and testing skills	

WEEKLY REPORT

WEEK–4 (From 08-01-2026to17-01-2026)

Objective of the Activity Done:

To develop the Flask backend and integrate the machine learning model for fraud prediction.

Detailed Report:

During this week, the team focused on developing the backend of the Fraud Detection System using Flask. The project structure was set up, and routes were created to handle user requests and process transaction data. This phase helped in understanding how backend systems manage data flow and interact with the frontend.

The machine learning model developed in the previous week was successfully integrated into the Flask application. Input forms were designed to collect transaction details from users, and prediction logic was implemented to process this data and generate results. This enabled the system to provide real- time fraud detection through a web interface.

By the end of the week, the application was tested to ensure proper functionality and accuracy of predictions. Bugs were identified and resolved to improve system performance. The integration of backend and machine learning components marked a significant step toward building a complete and functional fraud detection system.

ACTIVITY LOG FOR THE FIFTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 19-01-2026	Built user dashboard interface for transaction history	Learned data mapping and displaying historical records	
Day-2 20-01-2026	Implemented view for past and upcoming transaction alerts	Understood stateful component management	
Day-3 21-01-2026	Added functionality to mark/flag suspicious transactions	Learned conditional UI flows and dynamic UI updates	
Day-4 22-01-2026	Started admin panel for managing user accounts and data	Understood CRUD operations in Flask	
Day-5 23-01-2026	Worked on real-time transaction monitoring interface	Learned real-time data APIs	
Day-6 24-01-2026	Bug fixing and sprint review	Improved debugging and review techniques	

WEEKLY REPORT

WEEK–5 (From 19-01-2026to24-01-2026)

Objective of the Activity Done:

To develop the user dashboard and admin panel features, including transaction history and real-time monitoring.

Detailed Report:

During this week, the team focused on building the user dashboard to display transaction history and alerts for potential fraud. Components were developed to show past and upcoming transactions, and logic was implemented to highlight suspicious activity. This helped in understanding dynamic UI updates and state management in the application.

In parallel, work began on the admin panel, enabling CRUD operations for managing user accounts and transaction records. Real-time monitoring interfaces were also integrated, allowing the admin to track transactions as they occur. This phase provided hands-on experience with API integration and data handling in Flask.

By the end of the week, all features were tested, bugs were fixed, and a sprint review was conducted. The user dashboard and admin panel were fully functional, providing both users and admins with clear, actionable insights. Collaborative debugging and review sessions strengthened the team’s skills in maintaining code quality and improving application reliability.

ACTIVITY LOG FOR THE SIXTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 26-01-2026	Solution architecture design and documentation	Learned system architecture practices	
Day-2 27-01-2026	Started testing with test case writing	Understood QA process and scenario coverage	
Day-3 28-01-2026	User Acceptance Testing: registration/login	Gained practical UAT skills	
Day-4 29-01-2026	Bug tracking and reporting	Improved skills in bug reporting	
Day-5 30-01-2026	Performance and load testing (Lighthouse)	Learned web performance analysis	
Day-6 31-01-2026	Solution architecture review	Enhanced documentation and review skills	

WEEKLY REPORT

WEEK–6 (From 26-01-2026 to 31-01-2026)

Objective of the Activity Done:

Design solution architecture and ensure performance and reliability of the Fraud Detection System.

Detailed Report:

In Week 6, the team focused on finalizing the solution architecture and documenting the system structure. This included detailing modules, data flow, and integration points between the Flask backend and machine learning components. The documentation ensured clarity for future maintenance and collaboration.

Testing activities were then carried out, starting with writing test cases for core features such as registration, login, and transaction input. User Acceptance Testing (UAT) was conducted to simulate real user interactions and validate system functionality. Bugs identified during testing were tracked, reported, and resolved to improve application stability.

The team also performed performance and load testing using Lighthouse to assess web responsiveness and speed. The week concluded with a solution architecture review to ensure the system met design standards and was scalable. These activities strengthened the team's skills in QA, performance analysis, and technical documentation, preparing the project for final deployment.

ACTIVITY LOG FOR THE SEVENTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 02-02-2026	Deployment planning for Render	Learned about deployment strategies	
Day-2 03-02-2026	Configured deployment pipeline	Gained experience in CI/CD and server setup	
Day-3 04-02-2026	Demo video recording and presentation preparation	Improved communication and demonstration skills	
Day-4 05-02-2026	Implemented feedback from UAT	Enhanced iterative development and user-focused improvements	
Day-5 06-02-2026	Addressed responsive design and mobile optimization issues	Gained skills in responsive and adaptive UI design	
Day-6 07-02-2026	Final deployment and showcase	Learned final deployment best practices	

WEEKLY REPORT

WEEK–7 (From 02-02-2026 to07-02-2026)

Objective of the Activity Done:

Prepare and deploy the application, create documentation and demos.

Detailed Report:

During Week 7, the team focused on deploying the Fraud Detection Flask application. Deployment planning was done with Render, setting up the server environment and CI/CD pipeline to ensure smooth integration and live hosting. This phase provided practical experience in cloud deployment and configuration.

In parallel, a demo video was recorded, and presentation materials were prepared to showcase the project workflow and functionalities. Feedback from User Acceptance Testing (UAT) was implemented to improve the system's usability and responsiveness. Mobile optimization issues were also addressed, ensuring the application works effectively across devices.

By the end of the week, the project was successfully deployed on Render and showcased. The team gained valuable experience in deployment strategies, CI/CD practices, responsive design, and presenting a live, functional application. This week marked the transition of the project from development to a fully operational system.

ACTIVITY LOG FOR THE EIGHT WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 09-02-2026	Project documentation and codebase review	Learned report writing and code review	
Day-2 10-02-2026	Compiled final internship report	Enhanced skills in technical writing	
Day-3 11-02-2026	Prepared appendix: source code and video links	Understood comprehensive report compilation	
Day-4 12-02-2026	Final presentation preparations and rehearsal	Gained confidence in public speaking	
Day-5 13-02-2026	Delivered internship presentation	Improved skills presenting to technical audience	
Day-6 14-02-2026	Received feedback and reflected on internship	Learned to incorporate constructive criticism	

WEEK–8 (From 09-02-2026 to 14-02-2026)

Objective of the Activity Done:

Document final project, prepare presentation, and reflect on learning.

Detailed Report:

During Week 8, the focus shifted towards the completion of the internship journey by finalizing and compiling all components of the TransLingua project. A major part of the week involved preparing detailed technical documentation, including system architecture, key features, user workflows, and implementation details. Screenshots and code references were organized systematically to demonstrate different modules of the application. Care was taken to clearly present design decisions, challenges faced, and solutions implemented during the development process.

Alongside documentation, significant effort was dedicated to preparing the final presentation. The team worked collaboratively to highlight the main features of the application, explain the workflow, and demonstrate core functionalities such as text input processing, language detection, and real-time translation. Emphasis was placed on presenting both individual contributions and team efforts, along with challenges faced and improvements made based on feedback. Repeated practice sessions helped in improving confidence and clarity in presenting technical concepts.

Reflection was also an important part of this phase. Time was spent reviewing feedback received from mentors and incorporating useful suggestions into both the documentation and the application. This process helped in identifying the skills developed during the internship, areas for improvement, and future learning goals. The final phase of the internship not only strengthened technical knowledge but also enhanced communication skills and readiness for future career opportunities.

ACTIVITY LOG FOR THE NINETH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 16-02-2026	Initiated advanced translation module backend design	Understood API structure and AI request handling	
Day-2 17-02-2026	Implemented input data collection and preprocessing	Improved data handling and validation techniques	
Day-3 18-02-2026	Added multi-language selection interface	Learned state management and dynamic UI updates in React	
Day-4 19-02-2026	Integrated AI translation services (testing phase)	Understood AI service integration workflow	
Day-5 20-02-2026	Built translation result processing and output logic	Learned automated response handling	
Day-6 21-02-2026	Testing translation module and bug fixing	Improved testing and debugging skills	

WEEK–9 (From 16-02-2026 to 21-02-2026)

Objective of the Activity Done:

Finalize project documentation, enhance application features, and reflect on learning.

Detailed Report:

During Week 9, the focus was on finalizing the TransLingua project by completing documentation and refining remaining features. Time was dedicated to organizing all project components, including system architecture, modules, and implementation details, ensuring the report clearly represented the entire development process. Minor improvements were made to the application based on earlier testing feedback, enhancing overall performance and user experience.

Reflection also formed an integral part of the week. Time was allocated to reviewing feedback received from mentors during earlier phases of the internship and integrating constructive suggestions into both the report and the application. This reflective exercise helped in identifying the skills developed throughout the internship, areas for improvement, and future learning goals in Artificial Intelligence and software development.

The closing activities and mentor interactions not only strengthened my technical understanding but also boosted confidence and readiness for future professional opportunities. The internship experience concluded with a sense of achievement and preparedness to work on real-world AI-based applications.

ACTIVITY LOG FOR THE TENTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 23-02-2026	Created detailed product backlog for TransLingua features	Learned backlog creation, feature planning, and prioritization	
Day-2 24-02-2026	Started Sprint-1: User input and translation module development	Understood sprint planning and task breakdown	
Day-3 25-02-2026	Developed core translation functionality	Gained hands-on experience in frontend and backend integration	
Day-4 26-02-2026	Implemented translation response and output handling	Learned about automated processing and system response	
Day-5 27-02-2026	Integrated AI APIs for translation services	Understood AI API integration and workflow	
Day-6 28-02-2026	Sprint review and testing of translation module	Improved skills in sprint review and quality assurance	

WEEK–10 (From 23-02-2026 to 28-02-2026)

Objective of the Activity Done:

Enhance core modules, improve user interaction, and refine system functionality.

Detailed Report:

The team collaborated to design and implement backend processes that efficiently handle user input and interact with AI services for translation. Attention was given to ensuring smooth input validation and meaningful feedback, allowing users to easily understand and correct errors during interaction. The core translation module was tested with various input scenarios to ensure accuracy and reliability, forming a strong base for advanced features.

As part of improving the overall user experience, additional enhancements were integrated into the workflow. The team focused on refining AI response handling and optimizing processing speed to deliver faster and more accurate translations. In parallel, support for multiple input methods such as text and voice was strengthened to make the application more flexible and user-friendly. These improvements not only enhanced usability but also increased the efficiency and performance of the system.

ACTIVITY LOG FOR THE ELEVENTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 02-03-2026	Built translation history view interface	Learned data mapping and managing user history	
Day-2 03-03-2026	Implemented past translations display	Understood stateful component design	
Day-3 04-03-2026	Added edit/re-translate functionality	Learned conditional UI flows and dynamic updates	
Day-4 05-03-2026	Started admin panel (manage languages/settings)	Understood CRUD operations	
Day-5 06-03-2026	Worked on real-time translation updates	Learned real-time data processing	
Day-6 07-03-2026	Bug fixing and sprint review	Improved debugging and review techniques	

WEEK–11 (From 02-03-2026 to 07-03-2026)

Objective of the Activity Done:

Finalize project documentation, prepare presentation, and reflect on learning.

Detailed Report:

During Week 11, the focus shifted towards the completion of the internship journey by finalizing and compiling all components of the TransLingua project. A major part of the week involved preparing detailed technical documentation, including system architecture, key features, user workflows, and implementation practices followed during development. Screenshots and code references were carefully organized to demonstrate different modules of the application. Special attention was given to explaining design decisions, technical challenges, and solutions implemented throughout the project.

Alongside documentation, significant effort was dedicated to preparing the final internship presentation. The team worked collaboratively to highlight the key features of the application, explain the workflow, and demonstrate core functionalities such as input processing, language detection, and real-time translation. Emphasis was placed on presenting both individual contributions and team efforts, along with challenges faced and improvements made based on feedback. By practicing the presentation multiple times, the team improved confidence and clarity in explaining technical concepts to mentors and evaluators.

ACTIVITY LOG FOR THE TWELTH WEEK

Day& Date	Brief description of the Daily activity	Learning Outcome	Person Incharge Signature
Day-1 09-03-2026	Project documentation and codebase review	Learned technical documentation and code review practices	
Day-2 10-03-2026	Compiled final internship report for TransLingua	Enhanced skills in technical writing and structuring reports	
Day-3 11-03-2026	Prepared appendix: source code and demo links	Understood complete project documentation process	
Day-4 12-03-2026	Final presentation preparation and rehearsal	Gained confidence in communication and presentation	
Day-5 13-03-2026	Delivered final project presentation	Improved skills in presenting to technical audience	
Day-6 14-03-2026	Received feedback and reflected on internship	Learned to analyze and apply constructive feedback	

WEEK–12 (From 09-03-2026 to 14-03-2026)

Objective of the Activity Done:

Finalize project documentation, complete presentation, and reflect on learning.

Detailed Report:

During Week 12, the focus shifted towards the completion of the internship journey by finalizing and compiling all components of the TransLingua project. A major part of the week involved comprehensive technical documentation, including system architecture, key features, user workflows, and coding practices followed during development. Screenshots and code references were carefully organized to demonstrate different modules of the application. Special attention was given to summarizing design decisions, technical challenges, and solutions implemented throughout the internship.

As part of enhancing the overall application experience, improvements were made to ensure smooth interaction and efficient processing. The team focused on refining input handling, improving translation accuracy, and optimizing response time for better performance. Additional enhancements were made to ensure the application was user-friendly and responsive across different devices.

By the end of the sprint, the application was stable, efficient, and ready for final presentation and evaluation. The sprint review highlighted the effectiveness of the implemented features and overall system performance. Collaborative efforts, proper documentation, and continuous feedback helped in building a well-structured and scalable AI-based application, marking a successful completion of the internship.

Skills Acquired

During During my internship at SmartBridge, I acquired a wide range of technical and analytical skills that have become foundational for my future career in Artificial Intelligence and Machine Learning. The hands-on experience, combined with teamwork and continuous mentorship, helped me understand real-world data-driven problem solving, model development, and deployment processes while strengthening my analytical thinking and technical confidence.

1. Machine Learning Model Development:

- I developed strong skills in building and training machine learning models, particularly using the XGBoost algorithm for classification tasks.
- I learned how to preprocess data, split datasets, train models, and evaluate their performance using appropriate metrics.

2. Data Analysis and Preprocessing:

- I gained practical experience in handling datasets using Pandas and NumPy.
- I learned techniques such as data cleaning, handling missing values, encoding categorical variables, and feature engineering to improve model performance.

3. Model Evaluation and Optimization:

- I understood how to evaluate models using metrics like accuracy, precision, recall, and F1-score.
- I also learned how to tune models and reduce errors such as false positives and false negatives for better prediction reliability.

4. Model Deployment (Flask Integration):

- I gained experience in integrating machine learning models into a web application using Flask.
- I learned how to build APIs, handle user input, and deliver real-time predictions through a simple interface.

5. Testing and Debugging:

- I learned to create test cases for validating system functionality and model predictions.
- I developed skills in debugging errors, improving model performance, and ensuring system reliability.

6. Version Control and Collaboration:

- Using Git and GitHub, I collaborated with my team effectively by managing code versions, tracking changes, and resolving conflicts.
- This helped me understand real-world development workflows and improved my teamwork and communication skills.

Challenges Overcome

The internship journey involved several technical challenges that strengthened my problem-solving abilities and deepened my understanding of real-world Artificial Intelligence and Machine Learning systems.

1. **Handling Imbalanced Dataset:**

   Fraud detection datasets are highly imbalanced, with very few fraudulent cases compared to genuine transactions. This made it difficult for the model to learn effectively. I addressed this by carefully selecting evaluation metrics like precision, recall, and F1-score instead of relying only on accuracy, and by tuning the XGBoost model to improve detection of minority classes.

2. **Data Preprocessing and Feature Engineering:**

   Preparing raw transaction data for model training was challenging due to missing values, categorical variables, and inconsistent formats. I overcame this by applying data cleaning techniques, encoding categorical features, and selecting relevant features to improve model performance and reliability.

3. **Model Optimization and Overfitting:**

   Ensuring that the model performed well on unseen data was a key challenge. Initially, there were issues with overfitting. Through hyperparameter tuning, proper train-test splitting, and iterative evaluation, I improved the model's generalization capability.

4. **Integrating Model with Flask Application:**

   Deploying the trained model into a Flask-based web application required careful handling of input formats and preprocessing steps during runtime. I ensured that the same preprocessing pipeline used during training was applied before prediction, which helped maintain consistency and accuracy.

5. **Debugging and Performance Issues:**

   During development, issues such as incorrect predictions, slow response times, and input validation errors occurred. These were resolved through systematic debugging, testing multiple scenarios, and optimizing both the model and backend logic, resulting in a stable and efficient real-time prediction system.

Conclusion

The SmartBridge internship was a transformative learning experience that provided me with in-depth exposure to the field of Artificial Intelligence and Machine Learning. From the initial ideation phase to model development, testing, and deployment, each stage contributed to strengthening my technical knowledge and practical understanding of building real-world ML solutions. Working on the Online Payments Fraud Detection System allowed me to apply theoretical concepts to a meaningful and impactful problem.

Beyond technical skills, I developed a strong foundation in analytical thinking, data-driven decision-making, and systematic problem-solving. Collaborating with teammates and receiving continuous guidance from mentors helped me improve my communication skills, adaptability, and ability to work effectively in a team environment. Regular feedback and iterative improvements ensured that I maintained quality and accuracy throughout the project.

The challenges encountered during the internship, such as handling imbalanced data, optimizing model performance, and integrating the model into a web application, became valuable learning opportunities. These experiences enhanced my resilience, critical thinking, and ability to approach complex problems with confidence and creativity.

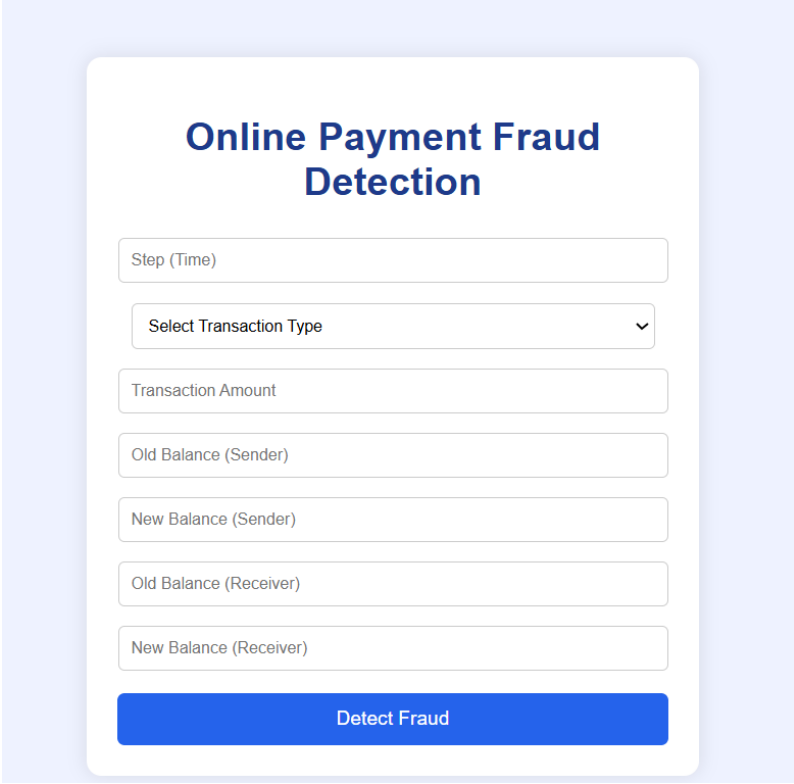
This internship has significantly prepared me to pursue a career in Artificial Intelligence and Machine Learning. I am grateful for the support, mentorship, and hands-on opportunities provided by SmartBridge, and I look forward to applying these skills to solve real-world problems and further advancing my expertise in AI/ML.

APPENDIX

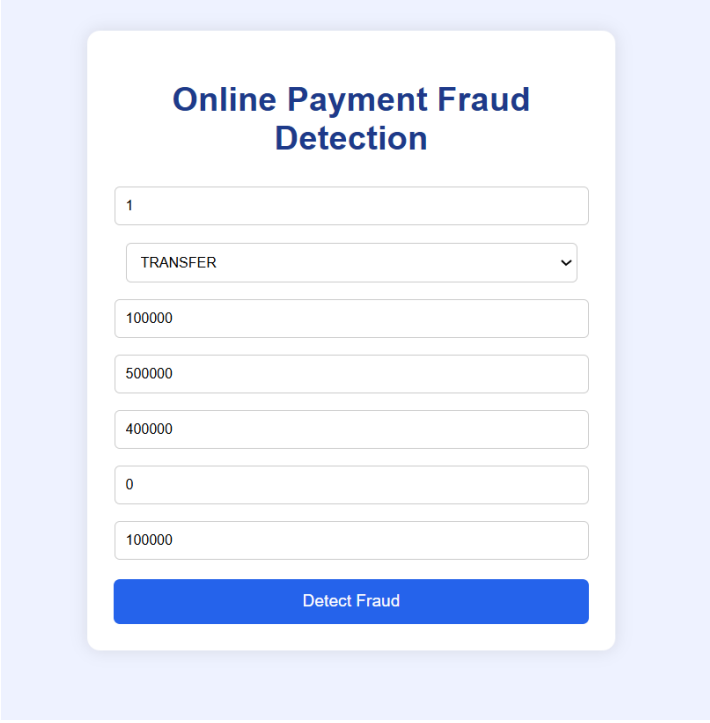
Source Code: https://github.com/Jaswanth-510/Fraud-Detection

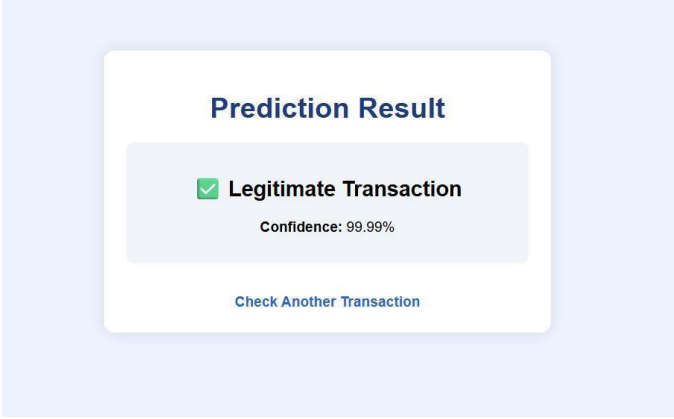
Results

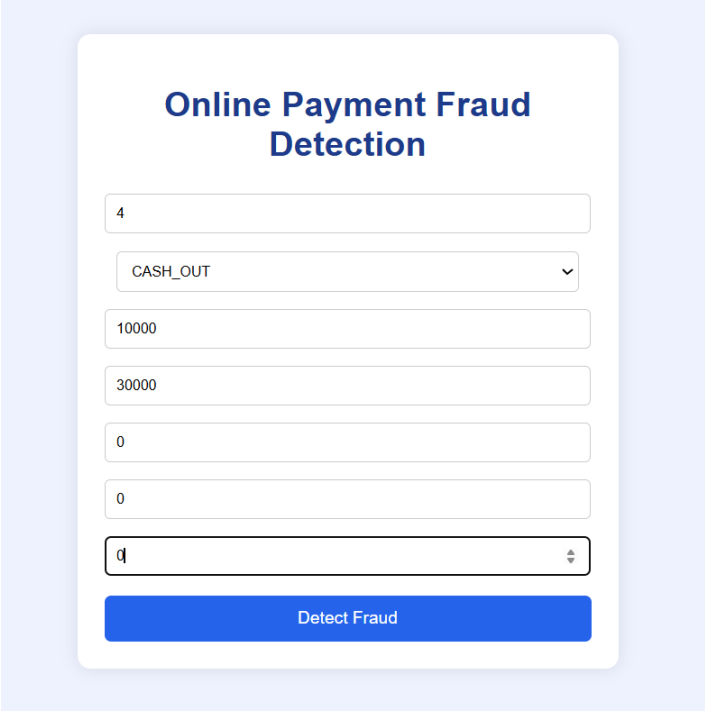
Homepage to enter inputs

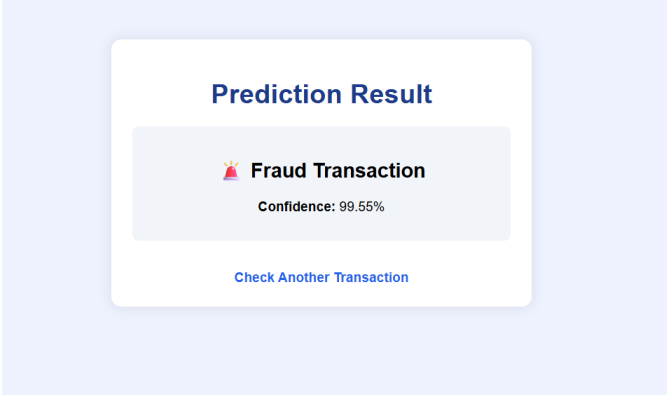


Entered the inputs to check




Prediction Result
Legitimate Transaction
Confidence: 99.99%
Check Another Transaction


Online Payment Fraud Detection
4
CASH_OUT
10000
30000
0
0
0
Detect Fraud


Prediction Result
Fraud Transaction
Confidence: 99.55%
Check Another Transaction